12.4 Volumes of Prisms and Cylinders

Name _____ Period ____ Date _____

Geometry 12.4 (30-minute plan) · Student sheet

I can…

- find the volume of a prism or a cylinder.
- find a missing dimension from a volume.
- use the scale factor of similar solids to compare their volumes.

Vocabulary · fill in the missing word

Volume — the number of cubic units _____ in the interior of a solid

Cavalieri's Principle — if two solids have the same height and the same _____ area at every level, then they have the same volume

Similar solids — two solids of the same type with equal ratios of _____ linear measures

Part 1 · Notes and examples (fill in each note, then work the example)

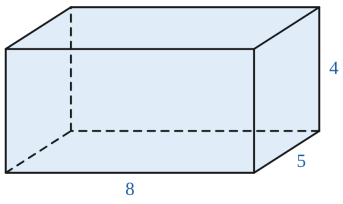
Volume of a prism: V = _____, where B is the area of a base and h is the height.

Example 1

Find the volume of each prism.

a. a rectangular prism 8 by 5 by 4 (shown)

b. a triangular prism with a right-triangle base with legs 6 and 8, and a length of 12

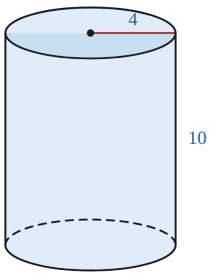


Volume of a cylinder: V = _____

Cavalieri's Principle: an oblique prism or cylinder has the _____ volume as a right one with the same base and height.

Example 2

Find the volume of the cylinder. Would an oblique cylinder with the same radius and height have a different volume? Round to the nearest hundredth.



Similar solids: if the scale factor is a : b, the volumes are in the ratio _____.

Example 3

Two cylinders are similar with a scale factor of 2 : 3. The smaller one has a volume of 80 cubic inches. Find the volume of the larger one.

Part 2 · Partner practice

Partner practice

A cylinder has a diameter of 6 and a height of 9. Find its volume. Round to the nearest hundredth.